MCS 320 Quiz 9 Friday 30 October 2026

Submit your Jupyter notebook with the answers to gradescope by 1:50pm.

1. The space curve defined $(x = \cos(t), y = \sin(t), z = \sin(t)^2)$, for $t \in [0, 2\pi]$, lies in the intersection of the cylinders $x^2 + y^2 = 1$ and $z = y^2$.

   For suitable ranges of the coordinates, draw the two cylinders in blue and green, and use red to draw the curve. Make the curve sufficiently thick so it is clearly visible.

2. Consider $p = x^2 + y^2 - 1$ and $q = y^2 - z$ in a polynomial ring with rational coefficients and pure lexicographic order $z > y > x$.

   (a) Compute a triangular form of the system defined by p and q.

   (b) Use the triangular form to derive a parametric representation for the curve in the intersection of the surfaces defined by p and q.

   (c) Make a plot to verify the correctness of the parametric representation, similar as in question 1.